Music streaming with "Listen Together" rooms: architecture

CLIENTS

iOS / Android apps
Player + prefetch cache. Sync engine:
clock offset, room anchor, drift correction

Web player
Same sync engine (Web Audio / MSE).
Adaptive-bitrate segment fetcher

What every client does in a room
Position = anchor + (server time - anchor time).
Reports sync error back as telemetry

REST / GraphQL

audio segments (HTTP/2, HTTP/3)

WSS: room state, commands

EDGE

API gateway / BFF
Auth (short-lived tokens), rate limits,
routing, request coalescing

Multi-CDN
Audio segments, artwork, cached metadata.
Origin shield + tiered cache

Realtime gateways (WebSocket)
Stateless, ~100 nodes for 2M sockets.
Subscribe to a room topic only if a local listener exists

subscribe / deliver

SERVICES

commands / state

Catalogue API
Tracks, albums,
artists. Cache-first
read path

Search + suggest API
Prefix from in-memory
FST, long tail from
OpenSearch

Playlist / user service
Playlists, library,
history reads

Playback service
Entitlement, signed
segment URLs,
DRM licences

Pub/Sub bus
NATS / Redis sharded
pub-sub. One topic
per active room

Room Service
Single-writer actor per
room. Holds a lease.
Stateful (~10k rooms/node)

origin fetch on cache miss

leases

services emit events

DATA AND MESSAGING

Catalogue DB
PostgreSQL shards
(~250 GB) + Redis
cache, CDN-cacheable

Search index
OpenSearch (~100M
docs) + popularity
signals

Playlist / user store
DynamoDB / Cassandra,
keyed by playlist id,
revision numbers

Object storage
Masters + renditions
(~1 PB). Immutable,
content-addressed

Leases + config
etcd. Room ownership
with fencing epochs

Room state store
Redis (replicated).
Room state + version;
async replica per region

CONTENT PIPELINE (offline path)

metadata + rights

renditions

Ingestion pipeline (idempotent workflow)
Label delivery (DDEX) -> validate and QC -> transcode (Opus/AAC ladder) -> package (CMAF, 2 s first segment) -> publish at embargo time

Release = flip a visible_at flag, after
CDN pre-warm of first segments.
Metadata -> outbox -> CDC -> search index.

EVENTS, ANALYTICS AND MONEY

Kafka event bus
Producers: playback, playlist, room service, ingestion. Events carry idempotency keys; producers use an outbox so nothing is lost if Kafka is slow

Listening history
Hot store (TTL) for
"recently played"

Data lake
S3 / Iceberg: analytics,
recommendations, ML

Royalty ledger
Immutable, reconciled with CDN
logs, daily payout batches

Sync telemetry
Client sync error, apply latency,
resyncs (ClickHouse + dashboards)